MTH2008 Real Analysis: Exercise Sheet 3

This Exercise Sheet covers the material presented in Weeks 7–10. By completing the exercises, you should develop an understanding of sequences and series, and Riemann integration.

1. Determine the value of $\lim_{n\to\infty} s_n$ for the following sequences. Be sure to justify your answers using the definition of a limit of a sequence.

(i) $(s_n)_{n\in\mathbb{N}}$ with $s_n = 1 + \dfrac{1}{n+1}$ for all $n \in \mathbb{N}$

(ii) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \dfrac{\sin(n)}{\sqrt{n}}$ for all $n \in \mathbb{N}$

(iii) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \dfrac{2n^2}{n^2+1}$ for all $n \in \mathbb{N}$

(iv) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \dfrac{2n^2}{1-n^2}$ for all $n \in \mathbb{N}$

2. Prove that the following sequences diverge, in particular prove if they diverge to $\pm\infty$ or oscillate.

(i) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \sqrt{\ln(n)}$ for all $n \in \mathbb{N}$

(ii) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \sin\left(\dfrac{n\pi}{2}\right)$ for all $n \in \mathbb{N}$

3. Find all L in the extended reals such that there exists a subsequence $(s_{n_k})_{k\in\mathbb{N}}$ of the given sequence $(s_n)_{n\in\mathbb{N}}$ with

$$\lim_{k\to\infty} s_{n_k} = L.$$

Moreover, for each such L, determine a subsequence $(s_{n_k})_{k\in\mathbb{N}}$ of $(s_n)_{n\in\mathbb{N}}$ such that $\lim_{k\to\infty} s_{n_k} = L$.

(i) $(s_n)_{n\in\mathbb{N}}$ with $s_n = (-1)^n \cdot n$ for all $n \in \mathbb{N}$

(ii) $(s_n)_{n\in\mathbb{N}}$ with $s_n = \left(1 + \dfrac{1}{n}\right) \cdot \cos\left(\dfrac{n\pi}{2}\right)$ for all $n \in \mathbb{N}$

4. Construct a sequence $(s_n)_{n\in\mathbb{N}}$ with the following property, or show that none exists: for each positive integer M, the sequence $(s_n)_{n\in\mathbb{N}}$ has a subsequence converging to M.

5. Let $(s_n)_{n\in\mathbb{N}}$ denote a sequence of real numbers.

(i) Suppose that $(s_n)_{n\in\mathbb{N}}$ is a Cauchy sequence, prove that it is bounded.

(ii) Suppose that $(s_n)_{n\in\mathbb{N}}$ is bounded, is it true that it is Cauchy?

6. Let $(s_n)_{n\in\mathbb{N}}$ and $(t_n)_{n\in\mathbb{N}}$ be Cauchy sequences of real numbers. Prove that the sequence $(s_n + t_n)_{n\in\mathbb{N}}$ is also a Cauchy sequence.

7. Let s_0 and s_1 be arbitrary fixed real numbers, and for $n \in \mathbb{N}$, set

$$s_{n+1} = \frac{s_n + s_{n-1}}{2}.$$

(i) If $s_0 = s_1$ show that $(s_n)_{n=0}^{\infty}$ is a constant sequence, and thus convergent.

(ii) In the case when $s_0 \neq s_1$, use Cauchy's convergence criterion to show that the sequence $(s_n)_{n=0}^{\infty}$ converges.

8. Use the Cauchy convergence criterion for series to show that the harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

9. Let $(a_n)_{n\in\mathbb{N}}$ denote a sequence of real numbers.

(i) Prove, if $\sum_{n\in\mathbb{N}} a_n$ is convergent, then for all $r \in \mathbb{N}$,

$$\lim_{n\to\infty} (a_n + a_{n+1} + \cdots + a_{n+r}) = 0.$$

(ii) Is the converse of Part (i) true? Justify your answer.

10. For each of the following sequences of functions $(f_n)_{n\in\mathbb{N}}$ determine the pointwise limit f of $(f_n)_{n\in\mathbb{N}}$, when it exists, and also determine if $(f_n)_{n\in\mathbb{N}}$ converges uniformly to f.

(i) $f_n : [0,1] \to \mathbb{R}$ where $f_n(x) = \dfrac{1}{1+nx}$

(ii) $f_n : \mathbb{R} \to \mathbb{R}$ where $f_n(x) = \dfrac{1}{n}\sin(nx)$

(iii) $f_n : \mathbb{R} \to \mathbb{R}$ where $f_n(x) = \begin{cases} 0 & \text{if } x \le n, \\ x-n & \text{if } x > n. \end{cases}$

(iv) $f_n : (0,1) \to \mathbb{R}$ where $f_n(x) = \begin{cases} n & \text{if } 0 < x \le \dfrac{1}{n}, \\ \dfrac{1}{x^4} & \text{if } \dfrac{1}{n} < x < 1. \end{cases}$

(v) $f_n : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ where $f_n(x) = \dfrac{e^x}{x^n}$

11. Let $a, b \in \mathbb{R}$ with $a < b$, and suppose that $(f_n)_{n\in\mathbb{N}}$ is a sequence of function, with $f_n : [a,b] \to \mathbb{R}$, for all $n \in \mathbb{N}$.

(i) Prove that if $(f_n)_{n\in\mathbb{N}}$ converges pointwise to some function $f : [a,b] \to \mathbb{R}$, and f_n is nondecreasing for each n, then f is nondecreasing.

(ii) Is Part (i) still true if we replace the word *nondecreasing* with *increasing*? Justify your answer.

12. Let $U \subseteq \mathbb{R}$ be a open set and let $(f_n)_{n\in\mathbb{N}}$ denote a sequence of bounded functions with $f_n : U \to \mathbb{R}$ for $n \in \mathbb{N}$. Namely, assume, for each $n \in \mathbb{N}$, there exists $M_n \in \mathbb{R}$ such that $|f_n(x)| \le M_n$ for all $x \in U$.

(i) Suppose that $(f_n)_{n\in\mathbb{N}}$ converges uniformly to a function $f : U \to \mathbb{N}$. Prove that f is bounded.

(ii) Suppose that (f_n) converges pointwise to a function $f : U \to \mathbb{R}$. Is it true that f is bounded? Justify your answer.

13. Is a uniform limit of differentiable functions differentiable? Namely, if $(f_n)_{n\in\mathbb{N}}$ is a sequence of differentiable functions converging uniformly to f, is f necessarily differentiable? Justify your answer.

14. Find the definite integrals $\int_a^b f(x)\,\mathrm{d}x$ of the following functions $f : \mathbb{R} \to \mathbb{R}$ using Riemann sums.

(i) $f(x) = x$ with $a = 1$ and $b = 3$

(ii) $f(x) = x^2 + x$ with $a = 0$ and $b = 1$

15. Use Riemann's criterion for integrability to show that the following functions $f : [a,b] \to \mathbb{R}$ are integrable.

(i) $f(x) = x^2$ with $a = 0$ and $b = 1$

(ii) $f(x) = \ln(x)$ with $a = 1$ and $b = 2$

16. Suppose that $f : [0,1] \to \mathbb{R}$ is monotonically decreasing and bounded. Prove that f is Riemann integrable on $[0,1]$.

17. Let $a, b \in \mathbb{R}$ with $a < b$. Prove that if $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$, is nonnegative, and is continuous at a point $x_0 \in (a, b)$ with $f(x_0) > 0$, then

$$\int_a^b f(x)\,\mathrm{d}x > 0.$$

18. Let $a, b \in \mathbb{R}$ with $a < b$, and let $f, g : [a, b] \to \mathbb{R}$ be Riemann integrable.

(i) Suppose that $f(x) > 0$ for all $x \in [a, b]$. Show that

$$\int_a^b f(x)\,\mathrm{d}x \geq 0.$$

(ii) Show that if $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x \leq \int_a^b g(x)\,\mathrm{d}x.$$

19. Let $a \in \mathbb{R}$. Find the derivative, with respect to x, of the following functions.

(i) $f : \mathbb{R} \to \mathbb{R}$ defined, for $x \in \mathbb{R}$, by

$$f(x) = \int_a^{x^3} \frac{1}{1 + \sin^2(t)}\,\mathrm{d}t$$

(ii) $g : \mathbb{R} \to \mathbb{R}$ defined, for $x \in \mathbb{R}$, by

$$g(x) = \int_{x^3}^{a} \frac{1}{1 + \sin^2(t)}\,\mathrm{d}t$$

(iii) $h : \mathbb{R} \to \mathbb{R}$ defined, for $x \in \mathbb{R}$, by

$$h(x) = \left(\int_a^{x} \frac{1}{1 + \sin^2(t)}\,\mathrm{d}t \right)^3$$

20. Suppose $h : \mathbb{R} \to \mathbb{R}$ is continuous, $f, g : \mathbb{R} \to \mathbb{R}$ are bounded and differentiable, and define $H : \mathbb{R} \to \mathbb{R}$, by

$$H(x) = \int_{f(x)}^{g(x)} h(t)\,\mathrm{d}t,$$

for $x \in \mathbb{R}$. Prove, for $x \in \mathbb{R}$, that $H'(x) = h(g(x))g'(x) - h(f(x))f'(x)$.

21. Determine the values of

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x \quad \text{and} \quad \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x$$

for the following sequences of real valued functions $(f_n)_{n\in\mathbb{N}}$ on $[0, 1]$.

(i) For $n \in \mathbb{N}$ and $x \in [0, 1]$, let $f_n(x) = \begin{cases} nx^{2n} & x \in [0, 1), \\ 0 & x = 1. \end{cases}$

(ii) For $n \in \mathbb{N}$ and $x \in [0, 1]$, let $f_n(x) = \dfrac{e^{-x^2/n} + x^3}{1 + \ln\left(\frac{x}{n} + 1\right)}$.

22. Find a sequence of Riemann integrable functions $(f_n)_{n\in\mathbb{N}}$ with $f_n : [0, 1] \to \mathbb{R}$, for all $n \in \mathbb{N}$, such that

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x$$

does not exist in $\mathbb{R}$, but

$$\int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x$$

does exist in $\mathbb{R}$.

23. Find a sequence of Riemann integrable functions $(f_n)_{n\in\mathbb{N}}$ with $f_n : [0,1] \to \mathbb{R}$, for all $n \in \mathbb{N}$, such that

$$\int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x$$

does not exist in $\mathbb{R}$, but

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x$$

does exist in $\mathbb{R}$.

24. Find a sequence of Riemann integrable functions $(f_n)_{n\in\mathbb{N}}$ with $f_n : [0,1] \to \mathbb{R}$, for all $n \in \mathbb{N}$, whose pointwise limit is not Riemann integrable.